

11.27 (a)  $c\rho$  and  $\vec{j}$  form a 4-vector. Therefore, in the reference frame  $K$ , the charge density and the current density are

$$c\rho = \gamma(c\rho' + \vec{\beta} \cdot \vec{j}') = \gamma c\rho' = c\rho', \quad \vec{j}_{||} = \gamma(\vec{j}'_{||} + \vec{\beta} c\rho') = \gamma\rho' c\vec{\beta} = \rho c\vec{\beta}, \text{ and } \vec{j}_{\perp} = 0,$$

to first order in  $\vec{\beta}$ . In the frame  $K$ , the magnetic dipole moment is

$$\vec{m} = \frac{1}{2} \int (\vec{r} \times \vec{j}) d^3x = \frac{1}{2} \left( \int \rho \vec{r} d^3x \right) \times c\vec{\beta}.$$

Since  $\rho d^3x = \rho' d^3x'$  is invariant and  $\vec{r}_{||} = \gamma(\vec{r}'_{||} + \beta ct')$ ,  $\vec{r}_{\perp} = \vec{r}'_{\perp}$  to first order, we have

$$\vec{m} = \frac{1}{2} \left( \int \rho' \vec{r} d^3x' \right) \times c\vec{\beta} = \frac{1}{2} \vec{p} \times c\vec{\beta}. \text{ This also shows that the electric dipole moment in the frame } K \text{ to first order in } \beta \text{ is still } \vec{p}.$$

$$(b) \text{ Again, } c\rho = \gamma(c\rho' + \vec{\beta} \cdot \vec{j}') = \gamma\vec{\beta} \cdot \vec{j}' = \vec{\beta} \cdot \vec{j}', \quad \vec{j}_{||} = \gamma(\vec{j}'_{||} + \vec{\beta} c\rho') = \gamma\vec{j}'_{||} = \vec{j}'_{||},$$

$$\vec{j}_{\perp} = \vec{j}'_{\perp}, \text{ to first order in } \beta. \text{ In frame } K,$$

$$\vec{p} = \int \rho \vec{r} d^3x$$

By the same argument as in part (a),  $\vec{r} = \vec{r}'$ ,  $d^3x = d^3x'$ , then

$$c\vec{p} = \int (\vec{\beta} \cdot \vec{j}') \vec{r} d^3x'$$

Following the same argument as in section 5.6

$$\begin{aligned} c p_i &= \beta_j \int J_j' x_i d^3x' = \frac{1}{2} \beta_j \int (J_j' x_i - J_i' x_j) d^3x' = \frac{1}{2} \beta_j \epsilon_{jik} \int (\vec{j}' \times \vec{r})_k d^3x' \\ &= \frac{1}{2} \epsilon_{ijk} \beta_i \int (\vec{r} \times \vec{j}')_k d^3x' = \left[ \vec{\beta} \times \left( \frac{1}{2} \int (\vec{r} \times \vec{j}') d^3x' \right) \right]_i = (\vec{\beta} \times \vec{m})_i, \end{aligned}$$

where we have used the definition of magnetic dipole moment,  $\vec{m} = \frac{1}{2} \int (\vec{r} \times \vec{j}) d^3x$ .

Therefore,  $\vec{p} = \vec{\beta} \times \vec{m} / c$ .